Sunset endpoint refinement, part 1: the intensity-dressed cubic coupling and the single missing constant

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

After U6 the SC-SCOPE lift's endpoint problem is the sunset channel alone (conservative ratio $\times 0.97$ at $I = 2 \times 10^{-3}$). This note delivers the first refinement: the intensity-dressed coupling, which the U4 table conservatively froze at the production-anchor value. Scope: estimate-grade until the single named constant below is machine-certified.

2. The intensity-dressed coupling

The U4 bound used $u_{\rm eff}(M_R)$ with $M_R = M(m^*) = 0.109414$ at the production dressing $m^* = 0.3045$. But the sunset of a competitor at intensity I is evaluated at ITS dressed kernel,

$$\hat r(I) \;=\; r_R + 2\lambda' I: \qquad \hat r(2{\times}10^{-3}) \;=\; 0.30453 + 2 \times 8.0551 \times 0.002 \;=\; 0.33675,$$

and the coupling dressing uses $M(\hat r(I))$, which is STRICTLY BELOW M_R by the monotonicity of M (unit U2, L1). Linear-response estimate via the EXACT identity $M'(m)\big|_{m=\hat r} = -J(0)$ (both sides are $-\int [\,\hat r + C(k^2 - q_0^2)^2\,]^{-2}$; $J(0) = 0.290$ is certified at the chain's dressed kernel):

$$M(0.33675) \;\approx\; M_R - 0.290 \times (0.33675 - 0.30453) \;=\; 0.10941 - 0.00934 \;=\; 0.10007,$$

$$u_{\rm eff} \;\approx\; -0.86 + 32.4 \times 0.10007 \;=\; 2.382, \qquad u_{\rm eff}^2 = 5.675 \quad (-21\% \text{ vs U4}).$$

The endpoint sunset bound becomes

$$\tfrac{1}{2}\, u_{\rm eff}^2\,(2I) \cdot 6J(0)\, M(0.33675) \;=\; \tfrac{1}{2} \times 5.675 \times 0.004 \times 6 \times 0.290 \times 0.10007 \;=\; 1.977 \times 10^{-3},$$

against the composed endpoint margin 2.658×10^{-3}:

$$\boxed{\text{ratio} \;\approx\; \times 1.34 \quad (\text{vs } \;\times 0.97 \text{ in U4})}$$

— POSITIVE, i.e. the conservative endpoint failure of U4 is removed by the correct intensity dressing, CONDITIONAL on the single constant below.

The named missing constant (M-ENDPOINT). M is CONVEX ($M''(m) = 2\int [m + D]^{-3} > 0$), so the linearization UNDER-estimates $M(0.33675)$ and hence UNDER-estimates $u_{\rm eff}$: the $\times 1.34$ is an estimate, not a proven floor. The proven version requires ONE new machine number: $M(0.33675)$ evaluated with a one-sided convergence check (a single quadrature; the convex/secant sandwich $M_R - 0.290\,\Delta \le M(0.33675) \le M_R$ brackets it a priori in $[0.1001, 0.1094]$, and the bound is monotone in it — the WORST case $M = M_R$ reproduces U4's $\times 0.97$, the estimate 0.1001 gives $\times 1.34$). Registered as M-ENDPOINT with the follow-up script.

3. Honest negative finding: the kernel axis

U4 listed lift input (a) "per-transfer $\widehat{G^3}$ decay." At the dominant transfers $|t| = q_0$ (the shell modes carrying $\hat{g}_3$'s mass), the sunset kernel $\widehat{G^3}(t) = \iint G(k_1)G(k_2)G(t-k_1-k_2)$ is supported on configurations with all three internal momenta near the shell — and three on-shell momenta CAN sum to an on-shell vector (the 4-wave resonance geometry of the production N_4 accounting). The kernel at $|t| = q_0$ is therefore NOT parametrically below its Young bound $J(0)\,M$; the resonance phase space saturates a finite fraction of it. CONSEQUENCE (recorded so the programme does not chase a dead end): the kernel axis contributes at most an $O(1)$ factor; the lift's quantitative load is carried by the coupling axis (this note) and, if a further factor is needed, the counting axis (the $\hat{g}_3$ mass distribution over transfers — note that for a competitor with n readings the per-transfer weight is $u_{\text{eff}}^2\,2I/n$-scale while the J-weighted sum re-concentrates, so counting helps only for structured subfamilies; assessed LOW-YIELD).

4. State of the SC-SCOPE lift after U6 + U7

Channel / axis	$I = 4\times10^{-4}$	$I = 10^{-3}$	$I = 2\times10^{-3}$
tadpole (U6)		ABSENT identically (lemma sketch)	
sunset, U4 conservative	$\times7.7$	$\times2.8$	$\times0.97$
sunset, intensity-dressed (U7)	$\times7.7$ (unchanged)	$\sim\times3.2$ (est.)	$\sim\times1.34$ (est., M-ENDPOINT pending)

(The $I = 10^{-3}$ entry uses $\hat{r}(10^{-3}) = 0.3206$, $M \approx 0.1047$, $u_{\text{eff}}^2 \approx 6.18$: bound 1.20×10^{-3} vs 3.83×10^{-3}.) If M-ENDPOINT lands inside its convex/secant bracket, the third-cumulant landscape is positive at all three anchors at estimate grade — making the SC-SCOPE lift to third order a FEASIBLE registered programme (formal inputs remaining: R-U6-1/R-U6-2 from U6; M-ENDPOINT + the assembled third-order inequality from this note; the quartic-difference channel writeup from U4).

5. Numerical verification

NO machine numbers were executed this session (sandbox shell unavailable). New estimate-grade arithmetic on certified constants: $\hat{r}(I)$ values (0.33675, 0.3206), the linear-response M estimates (0.10007, 0.1047), the dressed couplings (2.382, 2.486), the bounds (1.977×10^{-3}, 1.20×10^{-3}), the ratios ($\times1.34$, $\times3.2$). All to be machine-asserted by the registered follow-up script (planned: codes/vacuum/ third_cumulant_assessment.py, shared with U4, plus the M-ENDPOINT quadrature with a one-sided refinement check).

Quantitative sanity check (mandatory): dimensional — $\hat{r}(I) = r_R + 2\lambda' I$ adds mass^2-like quantities consistently ($\lambda' I$ carries the dressing's units as everywhere in the chain); identity check — $M'(\hat{r}) = -J(0)$ is exact by construction (both are the same integral), so the linear-response step introduces no new kernel object; bracket check — the convex/secant sandwich $[0.1001, 0.1094]$ has width 9% of the central value and the endpoint ratio is monotone across it ($\times1.34 \to \times0.97$), so M-ENDPOINT decides the sign of the refinement claim cleanly — exactly why it is registered as THE missing constant; sign-direction — using the smaller M reduces BOTH the coupling and the kernel, both in the bound-shrinking direction, and both reductions are real physics (stronger dressing at higher intensity), not bookkeeping.

6. Devil's-advocate (three objections; CLAUDE.md 6.3)

(α) *"The composed margin denominator should also be re-evaluated at the endpoint dressing — maybe it shrinks too, cancelling the gain."* DISMISSED: the composed margin $0.00432(1-1/2.6)$ is built from the Prop-A floor (anchor closed forms, I-independent) and the AddE-hardened STEP-5B ratio at $I = 2\times10^{-3}$ — BOTH already evaluated at the endpoint's own dressed kernel in the certified chain. No constant in the denominator was frozen at the production anchor; the asymmetry the U7 refinement fixes existed only in the U4 numerator.

(β) *"The linearization uses $M'(\hat{r}) = -J(0)$ with $J(0)$ certified at ONE dressing; across $[0.3045, 0.3367]$ the slope varies, and convexity makes the estimate non-conservative."* VALID-with-mitigation — and this is the note's own central caveat: the $\times1.34$ is labelled estimate-grade everywhere, the convexity direction is stated, the a-priori bracket is supplied, and the proven version is reduced to ONE registered machine number (M-ENDPOINT) whose worst-bracket value reproduces U4's honest $\times0.97$. Nothing is claimed beyond the bracket.

(γ) *"The resonance argument in section 3 is qualitative; declaring the kernel axis low-yield without computing $\widehat{G^3}(q_0)$ might discard a real factor."* VALID-with-mitigation: the finding is registered as an honest-direction assessment (it can only make the programme MORE conservative — it discourages reliance on an unverified improvement); the M-ENDPOINT follow-up script can evaluate $\widehat{G^3}(q_0)$ alongside at negligible extra cost, and if a real factor exists it strengthens the lift beyond this note's claims.

7. Result footer

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Result ID:               B5-BEYOND-LAYER-BOUND / sunset-endpoint-
                         refinement (unit U7)
Precise statement:       REFINEMENT (T3, estimate-grade): the sunset
                         bound's coupling must be dressed at each
                         anchor's own r_hat(I) = r_R + 2 lam' I; via
                         the exact identity M'(r_hat) = -J(0), the
                         endpoint coupling drops u_eff^2 7.209 ->
                         ~5.675 and the kernel M-factor 0.1094 ->
                         ~0.1001, lifting the endpoint sunset ratio
                         x0.97 -> ~x1.34 (and x2.8 -> ~x3.2 at 1e-3).
                         CONDITIONAL on the named constant M-ENDPOINT
                         = M(0.33675), a-priori bracketed in
                         [0.1001, 0.1094]. HONEST NEGATIVE: the kernel
                         axis is low-yield (4-wave resonance phase
                         space saturates the Young bound at shell
                         transfers).
Scope:                   SC-SCOPE lift programme, sunset axis;
                         estimate-grade pending M-ENDPOINT.
Dependencies:            U4 (channel + arithmetic), U6 (tadpole
                         absent), U2 L1 (monotonicity/convexity of M),
                         certified constants (J(0), r_R, lam', M_R,
                         AddE margins).
Evidence grade:          ESTIMATE (T3 structure + T2 numbers); machine
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	asserts NOT executed (registered).
Reproduction command:	(registered follow-up) python codes/vacuum/third_cumulant_assessment.py (shared with U4; adds M-ENDPOINT quadrature + optional G^3(q0) evaluation)
Expected output:	M(0.33675) inside [0.1001, 0.1094] with one-sided refinement check; all U7 table entries asserted; JSON artefact under claims/B5-BEYOND-LAYER-BOUND/runs/.
Falsification gate:	M-ENDPOINT outside its convex/secant bracket (would indicate an error in the certified inputs or in L1); the assembled third-order inequality failing at any anchor after M-ENDPOINT lands (honest negative for the lift, pre-registered).
Tier before / after:	No tier action; B5 stays T5-CANDIDATE.
No-overclaim statement:	The x1.34 endpoint ratio is ESTIMATE-GRADE (convexity direction stated; worst-bracket reproduces x0.97). Nothing here lifts SC-SCOPE; the U1 theorem is untouched. PDF build + linter PENDING operator-side.
Next required action:	M-ENDPOINT machine evaluation + the shared U4/U7 assert script; then the assembled third-order inequality note (the actual lift candidate) if the operator directs it. Mainline continues (U8).